

Cybersecure and Cyber-Resilient European Cloud–Edge–IoT Computing Continuum: A Strategic Roadmap

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ABSTRACT The distribution of computation across cloud, edge, and IoT tiers, combined with the simultaneous operationalisation of NIS2, the Cyber Resilience Act, the AI Act, and DORA, has made cybersecurity, resilience, and digital sovereignty an integrated strategic problem for Europe. Perimeter-oriented security models do not scale to heterogeneous, physically exposed, and multi-jurisdictional continuum infrastructures. This paper addresses that problem by proposing a three-stage methodology for identifying and prioritising cybersecurity capabilities: architectural decomposition of the continuum into six Security Layers and four Transversal Security Cross-Layers together with their constituent capabilities (Stage 1); consolidation of those capabilities into priority clusters and analysis of each cluster through a seven-dimension template (Stage 2); and projection of the clusters onto phased transition matrices over a 2026–2040 horizon (Stage 3). Applying the methodology, the paper identifies a set of layer and cross-layer cybersecurity capabilities spanning hardware roots of trust, post-quantum cryptographic agility, lightweight edge security, federated interoperability, supply-chain verification, cognitive security operations, pan-European network federation, confidential computing, resilient infrastructure, and proactive threat management. These capabilities are consolidated into three executive priority clusters: Sovereign Trusted Infrastructure, Cognitive Security and Autonomous Cyber Defence, and Federated Operational Sovereignty, each analysed through the seven-dimension template and then projected over a 2026–2040 transition horizon. The contribution is both methodological, providing a reproducible template for cybersecurity roadmap construction, and strategic, framing autonomous, federated, and certified Security-by-Design as the operational target for European digital infrastructure.

INDEX TERMS Certification, Cloud–Edge–IoT continuum, confidential computing, cyber-resilience, digital sovereignty, European Union, post-quantum cryptography, security roadmap, Zero-Trust architecture.

I. INTRODUCTION

The shift from centralised cloud to a distributed Cloud–Edge–IoT continuum relocates computation, storage, and inference closer to devices, users, and physical processes [1]–[3]. The architectural gains (lower latency, higher operational autonomy, and locality of data processing) come at a structural cost: attack surface, system heterogeneity, and the number of administrative and trust domains that a defender must reason about simultaneously all increase sharply, while the perimeter-oriented and siloed security models designed for consolidated data-centre environments become structurally inadequate for infrastructures that are geographically dis-

persed, resource-constrained, and physically exposed [4]–[6]. Each relocation of workload across providers and jurisdictions is also a transfer of control, which makes continuum security inseparable from digital sovereignty, that is, the ability to govern, audit, and operate critical infrastructure under credible European influence. This coupling is sharpened by concurrent global trends, namely hyperscaler concentration in the EU cloud market [7], the migration to post-quantum cryptography [8], and the weaponisation of generative AI against both human operators and AI-enabled defences, and by the European regulatory response, which translates these pressures into binding obligations on essential entities, prod-

ucts with digital elements, financial services, and high-risk AI systems [9]–[12].

Europe faces this transition under four simultaneous regulatory activations. NIS2 [9], the Cyber Resilience Act (CRA) [10], the AI Act [12], and DORA [11] together raise the baseline of mandatory security, resilience, accountability, and supply chain obligations across essential entities, ICT products, high-risk AI systems, and financial sector third-party providers. The EU cybersecurity certification framework [13], [14] provides the assurance substrate these regulations presuppose. Operationalisation of this combined regime is uneven: national transposition lags the legal timetable in several large Member States [15], EUCS remains unadopted [16], and conformity assessment infrastructure for AI-specific security obligations is not yet in place [17].

The central argument of this paper is that Europe's normative leadership in trust, privacy, governance, and certification [18], [19] translates into strategic advantage only when it is embedded in deployable capabilities. We call the distance between normative leadership and deployable capacity the ambition–implementation gap. Closing this gap requires a transition from reactive, siloed security towards autonomous, federated, and certified Security-by-Design: infrastructures that sustain trustworthy operation under adverse conditions (autonomous), security mechanisms that span providers and jurisdictions (federated), and assurance embedded through the system lifecycle (certified Security-by-Design) [4], [20], [21].

In this paper, a roadmap is understood as a structured strategic instrument that links current capability gaps to future target states through phased actions, dependencies, and transition conditions. It is not a forecast of technological development, but a coordination tool for aligning research, innovation, policy, standardisation, deployment, and ecosystem investment over time [3]. The Cloud–Edge–IoT continuum refers to the distributed execution environment in which computation, storage, networking, sensing, and AI inference are dynamically placed across central cloud platforms, intermediate edge infrastructures, and resource-constrained devices [1]–[3]. The term continuum is used because these tiers are increasingly interdependent rather than isolated: workloads, data flows, trust decisions, and control functions move across them according to latency, resilience, privacy, energy, and sovereignty requirements [1], [22]. The term ecosystem captures the fact that secure operation depends not only on technologies, but also on providers, standards bodies, regulators, certification schemes, open-source communities, research infrastructures, SMEs, large industrial actors, and public authorities whose decisions jointly determine deployment capacity [3], [13], [14].

The paper makes five contributions.

- First, a three-stage methodology for constructing cybersecurity roadmaps under conditions of ecosystem uncertainty, combining architectural decomposition (Stage 1), priority clustering with seven-dimension analysis (Stage 2), and phased transition-matrix projection

(Stage 3), and integrating a Sovereignty Level construct as an interpretive strategic lens (Section III).

- Second, an architectural decomposition of the Cloud–Edge–IoT continuum into six Security Layers and four Transversal Security Cross-Layers, populated with their constituent capabilities, as the Stage 1 output (Section IV).
- Third, a consolidation of those capabilities into three priority clusters (Sovereign Trusted Infrastructure; Cognitive Security and Autonomous Cyber Defence; Federated Operational Sovereignty), each analysed through a seven-dimension template as the Stage 2 output (Section V).
- Fourth, a three-phase transition matrix for each priority cluster, projecting the seven-dimension analysis onto the 2026–2040 horizon and identifying, for each phase boundary, the Critical Enabler whose delivery determines whether progression to the next phase is plausible, as the Stage 3 output (Section VI).
- Fifth, a set of implementation-oriented recommendations derived from the cluster analysis and transition matrices, anchored in the current EU regulatory and industrial landscape, and focused on concrete Phase-1 Critical Enablers whose absence or delivery determines whether the 2040 target state is reachable (Section VII).

Section II provides the strategic and regulatory context. Section III presents the three-stage methodology. Section IV defines the layered system model and its constituent capabilities. Section V presents the three priority clusters and their seven-dimension analysis. Section VI presents the phase logic and transition matrices. Section VII states the recommendations. Section VIII discusses validation and limitations. Section IX concludes.

II. BACKGROUND AND DRIVING FACTORS

The first vector shaping the continuum transition is architectural: the continuum is characterised by decentralisation and physical exposure. Decentralisation complicates visibility, orchestration, patching, identity, and incident response across multiple administrative domains and increases the probability of inconsistent security postures across deployments [23], [24]. Physical exposure removes the environmental and access controls that hyperscale data centres rely on: edge nodes in factories, substations, transport hubs, and telecom sites are exposed to tampering, fault injection, side-channel attacks, and supply chain compromise [6]. Hardware roots of trust, secure boot, remote attestation, and tamper-aware design are therefore baseline requirements rather than optional enhancements [25], [26].

The European regulatory landscape constitutes a second major vector shaping the continuum transition. NIS2 [9], the CRA [10], the AI Act [12], DORA [11], the Data Act [27], and GDPR [28] together impose obligations that must be translated into architectures, assurance mechanisms, and lifecycle controls operating across cloud, edge, and IoT.

This simultaneity creates disproportionate implementation pressures for SMEs and smaller Member States.

Sovereignty is the third vector. Dependence on non-EU providers for secure silicon, cloud platforms, and enterprise security tooling creates geopolitical and operational risk. Extra-territorial data access under foreign jurisdiction has been confirmed as a practical constraint for EU users of hyperscaler services [29].

Europe's position in this landscape is asymmetric. A world-leading regulatory foundation and a rigorous certification framework [13], [14] coexist with structural industrial gaps. EU-headquartered vendors (Infineon, NXP, STMicroelectronics) design the dominant secure element products and achieved EUCC "High" certification for multiple product families in 2025 [30], but fabrication depends on non-EU foundries and designs remain proprietary, preventing sovereign auditability at the silicon level. Intel SGX/TDX, AMD SEV-SNP, and Arm TrustZone/CCA constitute the dominant operational hardware-backed TEE families across cloud/server and embedded deployments; EU-origin RISC-V alternatives remain largely at research or prototype stage [31]. *No EU-origin RISC-V processor has received EUCC or Common Criteria certification as of April 2026. The enterprise SIEM market is dominated by US platforms, with no EU-headquartered vendor in the Gartner Leaders quadrant as of late 2025 [32]. Hyperscalers hold approximately 70% of the EU cloud market; EU-headquartered providers collectively hold about 15%, stable since 2022 but below the 29% share recorded in 2017 [7]. For post-quantum migration the NIS Cooperation Group roadmap [8] now provides a binding framework, but production-scale post-quantum cryptographic (PQC) hardware tailored for edge and IoT is not yet available in Europe.*

III. METHODOLOGY

The methodology examines the heterogeneous landscape of the Cloud–Edge–IoT ecosystem and translates it into a structured roadmap for research, innovation, policy, deployment, and ecosystem development. It proceeds in three stages, followed by iterative review and refinement. Stage 1 (Section III-A) decomposes the continuum into architectural layers and cross-layers and identifies the specific capabilities required within each. Stage 2 (Section III-B) consolidates those capabilities into priority clusters and analyses each cluster through a seven-dimension template. Stage 3 (Section III-C) projects the priority clusters onto three-phase transition matrices covering the 2026–2040 horizon, identifying the Critical Enabler associated with each phase boundary.

This separation is deliberate. Stage 1 is a decompositional operation: it establishes the analytical vocabulary, exposes cross-layer dependencies, and prevents priorities from being selected as isolated technologies detached from their architectural context. Stage 2 is a synthetic operation: it aggregates capabilities that share dependencies or strategic outcomes into coherent investment and coordination domains and characterises each through a static seven-dimension

profile. Stage 3 is a projective operation: it embeds those static profiles in time, exposing the sequencing of research, regulation, deployment, and market uptake required to move each cluster from its 2026 baseline to its 2040 target. Critical Enablers, phase-gate conditions, and shared bottlenecks are products of Stage 3 rather than of Stage 2, because they only become visible once the cluster is examined as a temporal trajectory.

Rather than predicting when each technology or policy milestone will occur, the methodology organises the capabilities, dependencies, and transition conditions that make progress plausible. This distinction is important because the Cloud–Edge–IoT ecosystem evolves through the interaction of technical maturity, regulatory obligations, deployment readiness, market adoption, and industrial capacity. The methodology also links technical cybersecurity capabilities to broader European strategic objectives: resilience, interoperability, trust, and sovereignty.

In this paper, Sovereignty Level denotes the extent to which critical capabilities and dependencies can be shaped, governed, assured, and operated under European influence or control. It is assessed at the capability and dependency level in Stage 1, used to interpret the strategic significance of priority clusters in Stage 2, and projected as a per-phase qualitative band in Stage 3 (the SSL row of each transition matrix). It is used as an interpretive strategic lens rather than as a rigid quantitative score.

A. STAGE 1: ARCHITECTURAL DECOMPOSITION AND CAPABILITY IDENTIFICATION

Stage 1 decomposes the continuum into architectural vertical and cross-cutting layers. These layers are treated as analytically distinct but operationally interdependent: a vulnerability or governance gap at one layer typically propagates to adjacent layers, so effective security requires coordinated action across the vertical and horizontal extent of the architecture [6], [24]. For each layer, Stage 1 further identifies the constituent capabilities: the technical and operational sub-domains in which strategic investment, standardisation, and ecosystem building are required.

Stage 1 answers three questions:

- 1) Where are the major security and resilience problem domains within the continuum?
- 2) At which architectural layers do they emerge?
- 3) Which capabilities within each layer are sufficiently strategic, cross-cutting, or underdeveloped to justify dedicated roadmap attention?

Capabilities are derived from four sources: the architectural characteristics of distributed systems across HPC, cloud, edge, and device tiers; the recurring cybersecurity challenges observed at those tiers (hardware integrity, software supply chain verification, workload isolation, identity federation, data confidentiality, and perimeter control); cross-cutting strategic concerns (sovereignty, trust, interoperability, resilience, assurance, and regulatory alignment); and

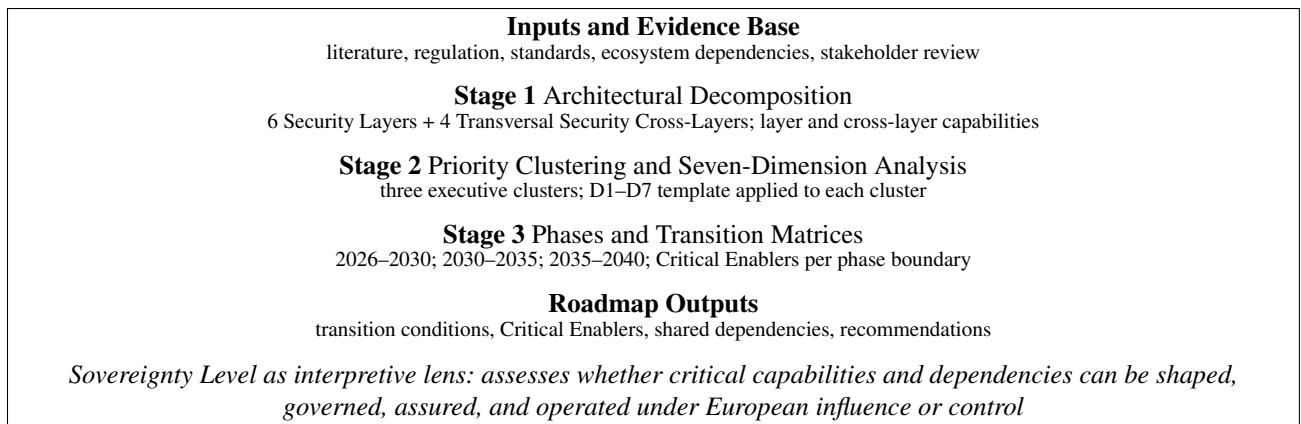


Figure 1. Three-stage methodology for constructing the cybersecurity and cyber-resilience roadmap. Stage 1 decomposes the Cloud–Edge–IoT continuum into Security Layers, Transversal Security Cross-Layers, and constituent capabilities. Stage 2 consolidates these capabilities into priority clusters and analyses each through a seven-dimension template. Stage 3 projects the clusters onto three-phase transition matrices, each row reflecting one of four assessment perspectives and the Critical Enabler row identifying the gate condition for phase progression. Sovereignty Level is used as an interpretive lens across capability, dependency, cluster, and phase assessment.

the need to connect technical architecture with operational constraints, governance requirements, and deployment pathways. A capability is therefore defined as a strategic ability that the European ecosystem should be able to design, deploy, govern, standardise, assure, coordinate, or scale in order to achieve trustworthy, resilient, and sovereign operation of the continuum. Capabilities may encompass technical controls, governance mechanisms, interoperability instruments, assurance functions, ecosystem coordination capacities, and deployment enablers.

The output of Stage 1 is a structured map of the problem space, organised into six Security Layers and four Transversal Security Cross-Layers, each populated with its constituent capabilities (see Section IV). Sovereignty-sensitive capabilities, where dependencies on external providers or assurance mechanisms are structurally significant, are identified so that Stage 2 can track their evolution.

B. STAGE 2: PRIORITY CLUSTERING AND SEVEN-DIMENSION ANALYSIS

Stage 2 groups the capabilities identified in Stage 1 into three Priority Clusters: (i) Sovereign Trusted Infrastructure, the hardware, cryptographic, software-update, data-protection, and resilience foundations that anchor trust across the continuum; (ii) Cognitive Security and Autonomous Cyber Defence, the AI-assisted operational layer that scales detection, analysis, and response beyond human-paced workflows; and (iii) Federated Operational Sovereignty, the interoperability, supply-chain verification, assurance, and federation substrate that converts individually sovereign components into a unified European trust fabric.

The use of three priority clusters is a deliberate synthesis choice rather than an assertion that only three cybersecurity priorities exist. This is consistent with technology-roadmapping practice, where roadmaps function as strategic planning and communication instruments that organise complex technical trajectories into a smaller number of decision-

relevant domains [33]. A roadmap may identify many relevant capabilities, but policy and funding coordination require higher-level priorities that communicate strategic direction, expose shared dependencies, and support portfolio-level decision-making. The three clusters therefore function as an executive structuring layer: they consolidate related capabilities into coherent investment and coordination domains while preserving the more granular capability analysis underneath.

Each priority cluster is analysed through a seven-dimension template applied consistently across the three clusters. The same template also preserves traceability to the underlying capabilities from which each cluster is derived:

- **(D1) Baseline**, describing the current state of the priority cluster, its capability maturity, dependencies, and main implementation gaps;
- **(D2) High-Impact Activities and Target Outcomes**, identifying the concrete research, innovation, standardisation, or deployment actions expected to advance the priority cluster, with each action assigned to a target phase and linked to the intended target outcome;
- **(D3) Primary Drivers**, identifying the threat and regulatory drivers that make the priority cluster urgent and non-deferrable;
- **(D4) Catalysts and Enablers**, identifying the funding instruments, regulatory and standardisation mechanisms, pilot infrastructures, and ecosystem coordination mechanisms that can accelerate the priority cluster;
- **(D5) Disruptors and Systemic Risks**, identifying technical cliffs such as the emergence of cryptographically relevant quantum computers, dependency risks such as supply-chain exposure and vendor lock-in, market concentration, and geopolitical constraints that could delay, weaken, or redirect the priority cluster;
- **(D6) Failsafe Mechanisms**, specifying fallback, isolation, rollback, redundancy, and graceful-degradation mechanisms that preserve security and resilience if the

primary capability or deployment pathway is delayed, compromised, or only partially realised; and

- **(D7) Transversal Meta-Layer**, assessing how the priority cluster contributes to broader European strategic objectives, including sovereignty, trust and ethics, and sustainable security.

D2 lists the activities that should advance a cluster but treats them as a static portfolio. D4 identifies the broader funding, standardisation, and pilot infrastructures that can accelerate the cluster as a whole. The temporal sequencing of these activities and the identification of the specific gate conditions whose delivery determines phase progression are operations of Stage 3, not Stage 2. The seven-dimension template therefore provides a structured and sufficiently comprehensive static profile of each priority cluster, enabling cross-cluster comparison, funding prioritisation, and the identification of shared dependencies, while leaving the temporal projection to the next stage.

C. STAGE 3: PHASE LOGIC AND TRANSITION MATRICES

Stage 3 projects each priority cluster onto a three-phase transition matrix covering the 2026–2040 horizon. The phases are:

- **Phase 1: Near-Term Implementation and Pilot Actions (2026–2030)**, focused on deploying near-ready solutions, enforcing enacted regulation, validating early pilots, and closing immediate governance gaps;
- **Phase 2: Mid-Term Scale-Up, Industrial R&D and Standardisation (2030–2035)**, focused on industrialising sovereign capabilities, federating architectures, scaling validated pilots, and building the workforce and institutional infrastructure required for wider adoption; and
- **Phase 3: Long-Term Strategic Autonomy and Leadership (2035–2040)**, focused on achieving operational maturity, consolidating European leadership, advancing frontier research, and resolving long-term governance and oversight questions.

1) Assessment Perspectives

The content of each phase column is informed by four assessment perspectives. The four are chosen because they jointly capture the distinct ecosystem conditions that must align for a cluster to progress, while being narrow enough to remain analytically tractable. Technical readiness alone does not predict adoption; regulatory obligation alone does not produce deployment; deployment alone does not produce market viability. The four perspectives are therefore treated as concurrent rather than sequential:

- **R&I Maturity**, covering the technical maturity of the capabilities underlying the priority cluster, the maturity of their enabling dependencies, the availability of pilot deployments, and the readiness of the research and innovation ecosystem to support transition;

- **Policy, Regulation, and Standards**, covering the regulatory obligations that drive the priority cluster, the standards and certification mechanisms that support it, and the policy instruments that can accelerate or constrain adoption;
- **Operational Deployment and Adoption**, covering the availability of deployable solutions, the maturity of deployment pathways, and the readiness of users, operators, and infrastructure providers to adopt the priority cluster; and
- **Ecosystem Uptake and Strategic Impact**, covering the ecosystem conditions that shape adoption, the strategic significance of achieving the target state, and the consequences of delay or failure.

These four perspectives are not represented as separate rows in the compact transition matrices presented in Section VI, but are folded into the Operational State row of each matrix, which characterises the observable state of the cluster within each phase column across all four perspectives concurrently. The narrative cluster analyses in Section V expand each perspective in prose.

2) Operational Deployment Phase and Strategic Sovereignty Level Rows

Each transition matrix uses two scalar rows to position the cluster within each phase. The Operational Deployment Phase (ODP) row uses the fixed vocabulary *Applied R&D* → *Fragmented Piloting* → *Standardised Integration* → *Ubiquitous Operation*, and characterises how broadly and how operationally the cluster is deployed. The Strategic Sovereignty Level (SSL) row uses the fixed vocabulary *High Foreign Dependency* → *Emerging EU Ecosystem* → *Full Sovereign Autonomy*. The SSL row operationalises, at the cluster level and per phase, the same Sovereignty Level construct introduced as an interpretive lens in Section III: it is the per-phase expression of that lens, not an independent metric. Where a cluster contains capabilities at heterogeneous sovereignty positions (Cluster 3 in Phase 1 being the principal example), the SSL cell records that heterogeneity rather than smoothing it into a single band.

3) Critical Enabler Row

Each phase column terminates in a Critical Enabler cell. The Critical Enabler is a single named gate condition whose delivery is treated as necessary for the cluster to progress to the next phase. It is distinguished from the broader Catalysts and Enablers of Stage 2 (D4) in two respects. First, where D4 lists the funding instruments, standards bodies, and pilot infrastructures that can accelerate a cluster as a whole, the Critical Enabler names the single condition whose absence would block phase transition irrespective of the wider catalyst landscape. Second, where D4 entries are static contextual enablers, the Critical Enabler is an event with a phase-bound delivery expectation: a regulatory enforcement date, a standard finalisation, a fabrication-line milestone, or an equivalent named occurrence. The Critical Enabler is

Execution environments			
HPC	Cloud	Edge	Devices
L6: Network and Perimeter			
L5: Data-Centric (PQC, TEE, QKD, SMPC)			
L4: User and Orchestration (ZT, IAM, PAM)			
L3: Application (Workloads, APIs)			
L2: Software (OS, Middleware, OTA)			
L1: Hardware (RoT, PhySec)			
CL1: Cyber-Resilient SW/AI Lifecycle			
CL2: Dynamic Risk and SecOps			
CL3: Interoperability (Simpl, CAMARA, MIMs)			
CL4: Compliance-by-Design and Assurance			

Figure 2. Security framework: four execution environments, six Security Layers (vertical stack, L1–L6), and four Transversal Security Cross-Layers (CL1–CL4).

therefore the principal phase-gate instrument of the roadmap, and the assessment of whether it is on track, at risk, or blocked is the principal analytical output of Stage 3.

A transition condition expressed through the Critical Enabler row indicates that a technical basis has matured, that a policy or standardisation basis has emerged, that a deployment pathway has been validated, that a strategic ecosystem condition has materialised, or that a Sovereignty Level improvement has become plausible through reduced dependency or increased European capacity. These conditions are not deterministic predictions or precise quantitative thresholds unless the available evidence supports such specificity.

IV. STAGE 1: CLOUD–EDGE–IOT CORE DIMENSIONS (LAYERS) AND CAPABILITIES IDENTIFICATION

The continuum is modelled as a layered reference architecture spanning four execution environments (HPC, Cloud, Edge, Devices) and organised into six *Security Layers* forming the vertical stack and four *Transversal Security Cross-Layers* acting as horizontal enablers, as shown in Figure 2. HPC provides large-scale scientific and AI training capacity; Cloud offers elastic service hosting and orchestration; Edge brings computation closer to users, industrial sites, telecom infrastructure, and data sources; and Devices include resource-constrained sensors, actuators, embedded systems, and IoT endpoints operating closest to the physical world. The decomposition aligns with established reference architectures, including ETSI MEC GS 003 [22], NIST SP 800-207 and SP 800-207A for Zero Trust [4], [5], NIST CSF 2.0 [34], ISO/IEC 27400 [35], and the ENISA baseline for IoT [36], and is consistent with the distributed computing-continuum framing in recent literature [1], [2] and in the European industrial technology roadmap for the next-generation cloud-edge [3].

The layers are analytically distinct but operationally inter-dependent: gaps at one layer can propagate to adjacent layers, as documented in the edge-computing security literature [6], [23], [24]. The model provides the shared vocabulary used throughout the paper and exposes cross-layer dependencies, allowing the priority clusters in Section V and the transition matrices in Section VI to be analysed as an interconnected

portfolio rather than as isolated priorities. The EU cybersecurity certification framework [13] and its EUCC operationalisation [14] provide the regulatory anchoring for several of the capabilities defined below.

The remainder of this section introduces each layer and cross-layer by describing its purpose within the security framework and listing its constituent capabilities. Table 1 consolidates the capabilities, their layer or cross-layer codes, and their principal references. Each capability is treated as an atomic analytical unit; the three priority clusters analysed in Section V are obtained by aggregating and translating these capabilities into actionable strategic needs.

A. SECURITY LAYERS (VERTICAL STACK)

The six Security Layers define the vertical stack of the continuum, moving from physical and silicon-level trust foundations through software, applications, identity and orchestration, data protection, and the network infrastructure. They are presented from the lowest trust anchor upwards because weaknesses at lower layers can undermine guarantees provided by higher layers, while higher-layer controls are required to operationalise security, governance, and resilience across distributed environments.

1) Layer 1: Hardware-Centric Security

Layer 1 addresses the physical and silicon-level trust foundations of the continuum and provides the verifiable anchors on which higher-layer security assurances depend. It covers hardware roots of trust, secure boot, remote attestation, Trusted Platform Modules, Physical Unclonable Functions, and physical-layer detection primitives [25], [26]. Without a verifiable silicon anchor, software-level attestation and cryptographic identity rest on assumptions that are difficult to validate in distributed and physically exposed environments. Layer 1 is the principal regulatory locus of hardware sovereignty in the continuum, and it is the layer at which EUCC “High” certification [14] operates.

Capability L1.1, Hardware Roots of Trust. Cryptographically verifiable device and supply-chain integrity from fabrication to deployment, with a strategic focus on advancing European open-hardware designs (RISC-V) toward EUCC “High” assurance. Technical foundations derive from TPM 2.0 [25] and integrity measurement architectures [26]. European instances include OpenTitan [37], Keystone [38], and the DARE SGA1 RISC-V initiative [39].

Capability L1.2, Physical-Layer Threat Detection and Fingerprinting. Exploitation of intrinsic wireless channel characteristics, including Channel State Information, Carrier Frequency Offset, and RSSI, to support real-time device authentication and anomaly detection at the lowest communication layer. This capability complements upper-layer cryptographic protections and provides additional identity assurance for resource-constrained devices that cannot reliably execute conventional authentication protocols [6], [24].

Table 1. Layer and cross-layer capabilities with their principal references.

Code	Capability	Primary references
L1.1	Hardware Roots of Trust; EU open-hardware (RISC-V) to EUCC “High”	[25], [26], [37]–[39]
L1.2	Physical-layer threat detection and RF fingerprinting	[6], [24]
L2.1	OS and middleware hardening; sovereign edge Linux	[4], [40]
L2.2	Secure OTA patch and update management	[41], [42]
L3.1	Workload security (service mesh, mTLS, sidecar)	[43], [44]
L3.2	In-band network telemetry and deep observability	[43], [44]
L3.3	API security gateway and policy enforcement	[40], [45]
L4.1	Zero Trust orchestration (PEP/PDP)	[4], [5]
L4.2	Federated IAM and SSI	[46]–[48]
L4.3	Privileged Access Management and JIT	[4], [5]
L5.1	Confidential computing and TEEs	[49]
L5.2	Post-Quantum Cryptography for edge/IoT	[8], [50]–[53]
L5.3	Signalling/data-plane cryptographic authentication	[54]–[57]
L5.4	Distributed reputation and trust networking (DLT)	[13], [58]–[60]
L5.5	QKD for critical links	[61], [62]
L5.6	Secure Multi-Party Computation	[12], [28], [49]
L6.1	Telco-cloud and O-RAN security	[54]–[57]
L6.2	Physical resilience and IT/OT safety	[6], [24], [54], [57]
L6.3	Edge-native autonomous guardians	[55]–[57]
L6.4	Non-terrestrial network and satellite security	[?], [63]
CL1.1	Continuous DevSecOps	[40]
CL1.2	Model integrity and DataBOMs	[64], [65]
CL1.3	Dynamic supply-chain verification; Runtime SBOMs	[66]
CL1.4	Automated adversarial testing and purple teaming	[67]
CL2.1	Automated SOC and collaborative CTI	[68]
CL2.2	High-precision telemetry and security signalling	[69], [70]
CL2.3	Internet of Behaviour and evidence sharing	[70]–[73]
CL2.4	AI-native analytics (NWDAF, LLMSecOps)	[64], [65], [69]
CL2.5	CTI lifecycle management	[70]–[73]
CL2.6	Deception technology and active defence	[70], [71]
CL3.1	Simpl integration and heterogeneous interoperability	[58]–[60]
CL3.2	Neutral host and multi-tenant isolation	[4], [5]
CL3.3	Pan-European NaaS federation	[74]
CL3.4	Semantic security interoperability	[59], [60]
CL3.5	Cyber range and digital twin federation	[3], [13], [68]
CL4.1	PQC readiness and Zero Trust architectures	[4], [5], [50]–[52]
CL4.2	Continuous automated certification (EUCC; EUCS)	[13], [14], [75], [76]
CL4.3	Multicloud Key Management as a Service	[4], [5], [50]–[52]
CL4.4	Privacy-preserving federated computation	[28], [49]
CL4.6	Cyber risk quantification and board-level reporting	[9], [11], [34]
CL4.7	AI trustworthiness and algorithmic accountability	[12], [64], [65]

2) Layer 2: Software-Centric Security

Layer 2 secures the software substrate on which continuum workloads execute, including hypervisors, container runtimes, distributed operating systems, and the pipelines that maintain them. Its central concerns are attack-surface reduction through OS and middleware hardening, and cryptographically verified, failure-resilient software distribution to field-deployed nodes for which physical access for remediation is limited or infeasible [41], [42]. Assurance at Layer 2 is necessary for trustworthy execution of the services and workloads built above it.

Capability L2.1, OS and Middleware Hardening. Security hardening of hypervisors (KVM/QEMU), container runtimes, and distributed operating systems, with emphasis on sovereign, lightweight Linux distributions optimised for edge resource constraints and reduced attack surface, in line with DevSecOps and Zero Trust guidance [4], [40].

Capability L2.2, Secure OTA Patch and Update Management. Cryptographically verified, failure-resilient over-

the-air update pipelines for field-deployed edge nodes such as smart grids, O-RAN radios, and industrial gateways. The IETF SUIT architecture [41] and the Uptane standard [42] provide the main reference frameworks.

3) Layer 3: Application-Centric Security

Layer 3 protects the workloads and service interfaces executing across the continuum, including containerised applications, serverless functions, service meshes, and the APIs that expose them. Application-layer defences are a key enforcement point for obligations that surface as runtime behaviours rather than infrastructure properties, including AI Act [12] and GDPR Article 25 [28] requirements. The NIST SP 800-204 microservices security series [40], [43], [44] and the OWASP API Security Top 10 [45] provide the main reference controls.

Capability L3.1, Workload Security. Securing containerised applications and serverless functions through hardened service meshes and sidecar proxies that provide mutual TLS,

traffic policy enforcement, and granular inspection while preserving operational performance, aligned with NIST SP 800-204 [43] and SP 800-204A [44].

Capability L3.2, In-Band Network Telemetry and Deep Observability. Embedding telemetry directly into packets or service-mesh flows traversing containerised workloads, enabling real-time security monitoring and anomaly detection without exclusive reliance on dedicated out-of-band monitoring infrastructure [43], [44].

Capability L3.3, API Security Gateway and Policy Enforcement. A unified enforcement point for authentication tokens, rate limiting, schema validation, and payload inspection, increasingly important as containerised and AI-enabled workloads communicate through REST and gRPC APIs. This capability follows the OWASP API Security Top 10 taxonomy [45] and the DevSecOps-for-service-mesh guidance in NIST SP 800-204C [40].

4) Layer 4: User and Orchestration-Centric Security

Layer 4 governs identity, access, and policy enforcement across the multi-provider continuum. It implements continuous, context-adaptive authorisation aligned with NIST SP 800-207 [4] and its cloud-native extension SP 800-207A [5]; federated and self-sovereign identity for humans, machines, and autonomous workloads based on the W3C Verifiable Credentials Data Model [46] and, in Europe, the eIDAS 2 framework [47] and its implementing acts for the EU Digital Identity Wallet [48]; and privileged-access controls that reduce reliance on standing credentials. Layer 4 is the layer at which identity-centric and policy-driven controls substitute for assumptions of stable network perimeter control.

Capability L4.1, Zero-Trust Orchestration. Policy Enforcement Point and Policy Decision Point architectures that enable continuous authorisation and dynamic access controls adapted to request context, device posture, workload identity, and operational risk, aligned with NIST SP 800-207 [4] and its cloud-native extension [5].

Capability L4.2, Federated Identity and Access Management. Federated and self-sovereign identity mechanisms enabling cryptographically verifiable, decentralised authentication of machines, users, and automated workloads across heterogeneous cloud-edge domains, based on the W3C Verifiable Credentials Data Model [46] and, in Europe, the eIDAS 2 framework [47] and the EU Digital Identity Wallet implementing acts [48].

Capability L4.3, Privileged Access Management and Just-In-Time Access. Reduction of over-provisioned standing privileges, a recurring enabler of advanced persistent threat campaigns, by replacing permanent access with time-bounded, contextually justified, and auditable grants, consistent with Zero Trust authorisation principles [4], [5].

5) Layer 5: Data-Centric Security

Layer 5 protects data across the continuum lifecycle: at rest, in transit, and in use. It spans confidential computing

through Trusted Execution Environments [49]; post-quantum cryptography based on NIST FIPS 203/204/205 [50]–[52]; signalling and data-plane protection; distributed reputation and trust; Quantum Key Distribution for selected high-value links [61], [62]; and Secure Multi-Party Computation for collaborative processing across mutually distrusting parties. Layer 5 is the layer at which GDPR Article 32 [28] and AI Act Article 15 [12] obligations on confidentiality, integrity, and robustness are most directly operationalised.

Capability L5.1, Confidential Computing. Protection of data in use through Trusted Execution Environments spanning the cloud-to-IoT data path, as defined by the Confidential Computing Consortium [49].

Capability L5.2, Post-Quantum Cryptography. Migration to quantum-resistant algorithms, namely ML-KEM [50], ML-DSA [51], and SLH-DSA [52], including adaptations for resource-constrained devices exposed to long-term “Harvest Now, Decrypt Later” threats. This capability is complemented by ETSI TS 103 744 hybrid key-exchange profiles [53] and aligned with the NIS CG PQC roadmap [8].

Capability L5.3, Signalling and Data-Plane Security. Embedding cryptographic authentication and integrity protection into signalling, control-plane, and data-plane interactions, beyond best-effort packet forwarding, in alignment with 3GPP TS 33.501 [54], O-RAN security specifications [55], [56], and the SNS JU SRIA [57].

Capability L5.4, Distributed Reputation and Trust Networking. Distributed reputation and trust mechanisms, potentially supported by Distributed Ledger Technology and smart contracts, enabling decentralised trust evaluation across heterogeneous providers without reliance on a single governance point, and aligned with Simpl federation interfaces [13], [58]–[60].

Capability L5.5, Quantum Key Distribution for Critical Links. Information-theoretically secure key exchange over optical fibre for selected point-to-point high-value links, such as government communications and critical infrastructure backbones, complementing algorithmic PQC for link-layer protection and governed by ETSI GS QKD 004 [62] and GS QKD 014 [61]. QKD is treated here as a complement to algorithmic PQC, not a substitute for it.

Capability L5.6, Secure Multi-Party Computation. Mechanisms allowing multiple mutually distrusting parties to jointly compute functions over combined private inputs without revealing those inputs, extending confidential computing (Capability L5.1) beyond single-party workloads to collaborative analytics across providers and jurisdictions, and supporting GDPR [28] and AI Act [12] obligations on data minimisation and confidentiality.

6) Layer 6: Network Infrastructure and Perimeter-Centric Security

Layer 6 secures the communication fabric of the continuum, including 5G and 6G cores, disaggregated RAN, IT/OT gateways, far-edge compute, and non-terrestrial network segments. It aligns with 3GPP TS 33.501 [54], O-RAN WG11

security specifications [55], Zero Trust for O-RAN [56], the SNS JU SRIA [57], and 3GPP NTN security [?], [63]. It is the layer at which cybersecurity intersects most directly with physical resilience, because failures or compromises in the communication fabric can cascade into operational disruption, safety hazards, or kinetic effects in critical infrastructures [6], [24].

Capability L6.1, Telco-Cloud and O-RAN Security. Hardening of distributed 5G and 6G cores and RAN components in alignment with 3GPP TS 33.501 [54], O-RAN WG11 security specifications [55], Zero Trust for O-RAN [56], and the SNS JU SRIA [57], with emphasis on secure network slice isolation, virtualised RAN protection, and sovereign operational control over critical telecom infrastructure.

Capability L6.2, Physical Resilience and Safety. IT/OT gateway security for high-availability edge environments such as smart grids, transport systems, and manufacturing sites, reducing the risk that digital breaches cascade into operational disruption, kinetic damage, or safety hazards [6], [24], [54], [57].

Capability L6.3, Edge-Native Autonomous Guardians. Lightweight, localised security control functions, including Security-as-a-Service aggregators, hosted directly at far-edge gateways or within Open-RAN Radio Intelligent Controllers (xApps/rApps) to enable autonomous enforcement at the network periphery [55]–[57].

Capability L6.4, Non-Terrestrial Network and Satellite Security. Security requirements for integrating satellite and non-terrestrial network segments into 6G architectures for coverage continuity, backhaul resilience, and emergency operations, aligned with 3GPP NTN specifications [?], [63] and the EU Space Programme.

B. TRANSVERSAL SECURITY CROSS-LAYERS

The four Transversal Security Cross-Layers define horizontal security capabilities that span the vertical stack and address properties no single layer can provide on its own: lifecycle integrity of software and AI components, operational response capacity, interoperability across providers and jurisdictions, and continuous regulatory assurance.

1) Cross-Layer 1: Automated Security Processes and Cyber-Resilient AI Lifecycle

Cross-Layer 1 spans the lifecycle of software and AI component security, from secure development and integration to runtime supply-chain verification and continuous adversarial validation. It treats AI systems both as defensive instruments and as attack targets, the latter framing being required by AI Act Article 9 risk-management obligations for high-risk systems [12]. CL1 aligns with service-mesh DevSecOps patterns [40], the NIST AI Risk Management Framework and its Generative AI Profile [64], [65], the SLSA provenance framework [66], and the adversarial robustness literature [67], with the objective of embedding cyber-resilience into how software and AI components are designed, built,

shipped, verified, and tested rather than treating assurance as a post-deployment activity.

Capability CL1.1, Continuous DevSecOps. AI-assisted code analysis, vulnerability detection, and remediation integrated into the earliest stages of the software lifecycle, following NIST SP 800-204C [40].

Capability CL1.2, Model Integrity and DataBOMs. Protection of AI training and deployment pipelines against data poisoning, unauthorised model manipulation, and insufficient traceability of training data and model artefacts, aligned with the NIST AI Risk Management Framework [64] and its Generative AI Profile [65], and supporting AI Act Article 9 risk-management requirements [12].

Capability CL1.3, Dynamic Supply-Chain Verification and Runtime SBOMs. Dynamic Software Bills of Materials and cryptographic proof-of-execution mechanisms that continuously verify the integrity of containerised microservices and AI components as they migrate across the multi-provider continuum, supported by the SLSA provenance framework [66].

Capability CL1.4, Automated Adversarial Testing and Continuous Purple Teaming. Continuous automated penetration testing, coverage-guided and protocol-level fuzzing, and purple-team automation against live or staging environments, providing the adversarial validation counterpart to DevSecOps and grounded in the adversarial robustness literature [67].

2) Cross-Layer 2: Dynamic Risk Management and Operational Cyber Resilience

Cross-Layer 2 governs threat detection, analytics, cyber threat intelligence, and operational response across the continuum. It supports SOC federation across Member States [68], integrates AI-native telemetry analytics via the 5G NWDAF architecture [69] and LLM-assisted security operations, and structures the cyber threat intelligence lifecycle using MITRE ATT&CK [71] and STIX/TAXII 2.1 [72], [73] as the structured-CTI foundation, complemented by ENISA threat landscape reporting [70]. Its role is to reduce detection-to-response timelines and improve coordinated operational resilience in environments where human-paced analysis alone is insufficient.

Capability CL2.1, Automated SOC and Collaborative Threat Intelligence. Federated SOC and collaborative threat-intelligence mechanisms across Member States, supported by legally protected sharing platforms that reduce institutional trust barriers and enable coordinated situational awareness [68].

Capability CL2.2, High-Precision Telemetry and Security Signalling. High-fidelity telemetry and security signalling mechanisms that feed real-time threat-intelligence and detection pipelines while managing bandwidth, latency, and alert-volume constraints, integrating with 5G NWDAF analytics interfaces [69] and threat-landscape baselines [70].

Capability CL2.3, Internet of Behaviour and Evidence Sharing. Collection, correlation, and sharing of be-

havioural evidence across infrastructure components to detect anomalous patterns invisible to purely signature-based defences, structured through MITRE ATT&CK [71] and STIX/TAXII 2.1 [72], [73] and informed by ENISA threat reporting [70].

Capability CL2.4, AI-Native Analytics Integration (NWDAF and LLMSecOps). Integration of cybersecurity telemetry with 5G NWDAF analytics [69] and LLM-assisted security operations to support triage, correlation, investigation, and response, framed by the NIST AI RMF and its Generative AI Profile [64], [65].

Capability CL2.5, Cyber Threat Intelligence Lifecycle Management. Governance of the CTI lifecycle as a prerequisite for federated SOC operation, using MITRE ATT&CK [71] and STIX/TAXII 2.1 [72], [73], complemented by ENISA threat landscape reporting [70].

Capability CL2.6, Deception Technology and Active Defence. Deployment of decoy assets, including honeypots, honeynets, credential canaries, and fake API endpoints, to generate high-fidelity alerts based on adversary interaction and to support earlier detection of lateral movement or reconnaissance [70], [71].

3) Cross-Layer 3: Interoperable Security Mechanisms and Open-Source Verifiability

Cross-Layer 3 enables secure interoperation across heterogeneous providers, infrastructures, and jurisdictions. It anchors federation in the Simpl open-source middleware for Common European Data Spaces [58], CAMARA Network APIs for pan-European compute federation [74], semantic interoperability through OASC Minimal Interoperability Mechanisms [59] and the ETSI NGSI-LD API [60], and federated cyber ranges for shared validation infrastructure. Its objective is to convert individually sovereign components into a collectively sovereign European trust fabric while avoiding new forms of vendor lock-in.

Capability CL3.1, Heterogeneous Interoperability and Simpl Integration. Alignment with the Simpl open-source smart middleware [58] to support secure federated access, policy portability, and interoperability across Common European Data Spaces, with semantic anchoring in OASC MIMs [59] and the ETSI NGSI-LD API [60].

Capability CL3.2, Neutral Host and Multi-Tenant Isolation. Enforcement of Zero Trust edge operations across network partitions, tenants, and shared-infrastructure models, ensuring that infrastructure sharing does not imply implicit trust between operators or workloads [4], [5].

Capability CL3.3, Pan-European NaaS Federation and Cross-Border Edge Roaming. Standardised interfaces leveraging CAMARA Network APIs [74] to support a federated Network-as-a-Service model for cross-border edge and connectivity services.

Capability CL3.4, Semantic Security Interoperability. Common ontologies and data models for security events, asset taxonomies, policy attributes, and risk classifications, op-

erationalised through OASC Minimal Interoperability Mechanisms [59] and the ETSI NGSI-LD API [60].

Capability CL3.5, Cyber Range and Digital Twin Federation. Federation of national cyber ranges and digital-twin environments into shared EU-wide validation infrastructure for testing cross-provider security, resilience, and interoperability assumptions, building on EU certification frameworks [13], CONCORDIA outputs [68], and the European industrial cloud-edge roadmap [3], and aligned with the Cyber Solidarity Act.

4) Cross-Layer 4: Compliance-by-Design, Agile Governance and Automated Assurance

Cross-Layer 4 embeds regulatory compliance, governance, and assurance considerations into technical architecture. It operationalises continuous assurance for cloud services (EUCS, when adopted) and edge products (EUCC) [13], [14] through machine-readable control catalogues and assessment results based on NIST OSCAL [75], [76]; cryptographic key management across multi-provider operations; privacy-preserving federated computation satisfying GDPR Article 25 [28]; cyber risk quantification that translates technical posture into financial and operational risk language under NIS2 [9], DORA [11], and NIST CSF 2.0 [34]; and AI trustworthiness and algorithmic accountability for autonomous security decisions under AI Act Article 9 [12], aligned with the NIST AI RMF [64]. Its objective is to shift assurance from point-in-time audit towards continuous, evidence-based, and telemetry-supported verification.

Capability CL4.1, Post-Quantum Readiness and Zero Trust Architectures. Continuous Zero Trust verification and post-quantum readiness embedded into cloud-edge infrastructure, aligned with NIST SP 800-207 [4] and SP 800-207A [5] and with FIPS 203/204/205 algorithm selection [50]–[52].

Capability CL4.2, Continuous Automated Certification. Lifecycle certification tracking for cloud services (EUCS, when adopted) and edge products (EUCC) [13], [14], operationalised through machine-readable control catalogues, assessment artefacts, and evidence exchange using NIST OSCAL [75], [76].

Capability CL4.3, Multicloud Key Management as a Service. Cryptographic key, policy, and lifecycle management across dynamic multi-provider computing-continuum operations, integrating Zero Trust authorisation [4], [5] with PQC algorithm selection [50]–[52].

Capability CL4.4, Privacy-Preserving Federated Computation. Cross-jurisdictional AI training and inference over distributed private datasets using privacy-preserving computation mechanisms [49], supporting GDPR Article 25 [28] data-protection-by-design objectives.

Capability CL4.6, Cyber Risk Quantification and Board-Level Reporting. Translation of technical security posture into financial, operational, and strategic risk language for decision-makers, aligned with NIS2 governance

obligations [9], DORA [11], and NIST CSF 2.0 governance functions [34].

Capability CL4.7, AI Trustworthiness and Algorithmic Accountability. Assurance that AI systems making or supporting autonomous security decisions remain robust, accountable, and auditable, in compliance with AI Act Article 9 [12] and aligned with the NIST AI RMF and Generative AI Profile [64], [65].

V. STAGE 2 OUTPUT: PRIORITY CLUSTERS AND SEVEN-DIMENSION ANALYSIS

Stage 2 consolidates the layer and cross-layer capabilities identified in Section IV into three coordinated priority clusters. Each cluster aggregates related capabilities that share architectural dependencies, enabling conditions, or strategic outcomes, and is presented below through the seven-dimension template defined in Section III (D1 Baseline; D2 High-Impact Activities and Target Outcomes; D3 Primary Drivers; D4 Catalysts and Enablers; D5 Disruptors and Systemic Risks; D6 Failsafe Mechanisms; D7 Transversal Meta-Layer). Where relevant, the analysis refers to the underlying layer or cross-layer capabilities from which the cluster is derived. The resulting priority clusters are subsequently projected through the three-phase transition matrices in Section VI.

A. CLUSTER 1: SOVEREIGN TRUSTED INFRASTRUCTURE

Cluster 1 consolidates the capabilities that provide the trusted execution substrate of the continuum: hardware roots of trust and physical-layer assurance (L1.1–L1.2), secure software foundations and update mechanisms (L2.1–L2.2), confidential computing and data-in-use protection (L5.1), post-quantum cryptographic agility and lightweight security for constrained environments (L5.2 and L5.5–L5.6), signalling and data-plane integrity (L5.3), and resilience mechanisms for critical network infrastructure (L6.1–L6.4). Federated policies and autonomous defence gain operational credibility in proportion to the maturity of this execution substrate. The substrate must support hardware integrity through verifiable silicon roots of trust, quantum-resistant confidentiality through crypto-agile schemes aligned with FIPS 203/204/205, and autonomous resilience under network or orchestration-plane compromise. The shared critical dependency is European sovereign fabrication capacity under the EU Chips Act [77], [78].

a: D1 Baseline.

At the hardware and physical layer (L1.1–L1.2), Infineon, NXP, and STMicroelectronics achieved EUCC “High” for proprietary secure element families in 2025 [30], but fabrication remains dependent on non-EU foundries and no EU-origin RISC-V processor has received EUCC or Common Criteria certification. At the software and update layer (L2.1–L2.2), field-deployed edge infrastructure remains dependent on heterogeneous hardening practices and update mecha-

nisms. At the data-centric layer (L5.1–L5.6), edge and IoT environments remain dependent on classical cryptography and non-European TEE families, while extreme-edge devices remain constrained in compute, energy, and memory. The NIS CG roadmap [8] requires national PQC roadmaps by December 2026, transition of high-risk critical infrastructure by December 2030, and full completion by December 2035. At the network infrastructure layer (L6.1–L6.4), current deployments often assume persistent connectivity and central orchestration.

b: D2 High-Impact Activities and Target Outcomes.

Phase 1 activities should establish the technical basis for the trusted execution substrate: fund three to five RISC-V secure element pilots under Horizon Europe Cluster 3 targeting EUCC “High” pre-evaluation and submit OpenTitan and Keystone prototypes for structured assessment (L1.1); evaluate physical-layer authentication and fingerprinting for constrained edge devices (L1.2); harden field-deployed edge software stacks and validate failure-resilient update paths (L2.1–L2.2); optimise FIPS 203/204/205 implementations for 8/16-bit microcontrollers and deploy hybrid ML-KEM plus ECDH-P384 on the cloud side according to ETSI TS 103 744 [53] (L5.2); and validate Island Mode soft-state protocols in 5G testbeds (L6.1–L6.4). Phase 2 activities should scale validated pilots into operational environments: bring the first EUCC “High” certified EU-manufactured RISC-V secure elements into production through Chips Act pilot lines such as NanoIC and FAMES (L1.1); deploy RISC-V Keystone TEEs in multi-provider Data Spaces (L5.1); certify 72 h Island Mode under 3GPP/ETSI fallback profiles in cross-border 5G corridors (L6.1–L6.4); and require hybrid PQC in DORA-scoped deployments (L5.2). Phase 3 activities should consolidate trusted infrastructure as the default operating model: make EU-designed, EUCC “High” certified secure elements available for at least $\geq 60\%$ of critical infrastructure edge nodes by 2040 (L1.1); support a quantum-resilient and crypto-agile continuum combining PQC, lightweight security mechanisms, confidential computing, QKD for selected critical links, and privacy-preserving collaborative computation (L5.1–L5.6); deploy zero-energy physical-layer authentication for battery-less mMTC devices (L1.2); and require validated ≥ 72 h autonomous Island Mode for NIS2-essential and DORA-critical deployments (L6.1–L6.4).

c: D3 Primary Drivers.

The primary threat drivers for this cluster arise from attacks against the trusted execution substrate itself: state-sponsored exploitation of debug interfaces and side-channel leakage in field-deployed nodes (L1.1–L1.2); supply-chain insertion and compromised update paths affecting software and firmware maintenance (L2.1–L2.2); “Harvest Now, Decrypt Later” campaigns targeting long-lived data confidentiality and cryptographic migration gaps (L5.1–L5.6); and cascading failures caused by cyberattacks, extreme

weather, or geopolitical disruption to communication and orchestration infrastructure (L6.1–L6.4). The regulatory drivers are equally distributed across the underlying capabilities: CRA Articles 10–11 establish cybersecurity and vulnerability-handling obligations with enforcement from December 2027 [10]; NIS2 Article 21(2)(e) requires policies and procedures for cryptography and encryption [9]; AI Act Articles 9 and 15 create risk-management, accuracy, robustness, and cybersecurity obligations for high-risk AI systems [12]; the NIS CG PQC roadmap defines the migration milestones for quantum-resistant cryptography [8]; DORA Article 11 requires ICT business continuity and disaster recovery capabilities [11]; and the SNS JU SRIA frames resilience as a core requirement for future 6G infrastructures [57].

d: D4 Catalysts and Enablers.

The main catalysts for this cluster map directly to the trusted-substrate capabilities. For hardware roots of trust and physical-layer assurance (L1.1–L1.2), the key enablers are the EU Chips Act pilot lines, including NanoIC, FAMES, APECS, PIXEurope, and WBG, with combined funding of approximately €3.7B [77]; the DARE SGA1 EuroHPC initiative, with €240M funding [39]; CEN/CENELEC JTC 13 hardware security standardisation; and EUCC “High” certification pathways. For secure software foundations and update mechanisms (L2.1–L2.2), relevant enablers include ECCC Strategic Agenda priorities [79], secure update frameworks, and coordinated pilot infrastructures for edge and critical-infrastructure deployments. For post-quantum cryptographic agility, confidential computing, QKD, and privacy-preserving computation (L5.1–L5.6), the primary enablers are NIST FIPS 203/204/205 [50]–[52], ETSI TS 103 744 hybrid key-exchange profiles [53], the NIS CG PQC roadmap, and confidential-computing and data-space pilots. For network resilience and Island Mode capabilities (L6.1–L6.4), SNS JU 6G resilience projects [57], 3GPP/ETSI standardisation, and cross-border 5G/6G testbeds provide the main transition infrastructure.

e: D5 Disruptors and Systemic Risks.

The main systemic risks affect different parts of the trusted execution substrate. For hardware roots of trust (L1.1), the key disruptor is the EU Chips Act fabrication-capacity shortfall: ECA Special Report 12/2025 projects approximately 11.7% global semiconductor share by 2030 against the 20% target [78], while the cancellation of Intel Magdeburg in July 2025 removed the largest EU-based advanced-node commitment. For physical-layer assurance and constrained edge security (L1.2), energy, memory, and signal-quality constraints may limit deployability of lightweight authentication and fingerprinting techniques. For secure software foundations and update mechanisms (L2.1–L2.2), the main risks are heterogeneous legacy stacks, incomplete asset inventories, and insufficiently resilient update paths in OT and field-deployed edge environments. For data-centric se-

curity (L5.1–L5.6), early arrival of a Cryptographically Relevant Quantum Computer before migration is complete, PQC computational overhead on battery-powered IoT devices, side-channel breakthroughs against current TEE architectures, and incomplete cryptographic inventories of legacy OT/IoT systems could delay transition. For network resilience (L6.1–L6.4), dependencies on central orchestration, telecom supply-chain concentration, and insufficiently validated fallback modes could weaken Island Mode readiness.

f: D6 Failsafe Mechanisms.

Failsafe mechanisms should preserve security and continuity when parts of the trusted substrate are delayed, compromised, or only partially available. For hardware and physical-layer capabilities (L1.1–L1.2), these include redundant trust anchors, layered lightweight cryptography, PHY signatures as last-resort authentication under energy starvation, and tamper-resistant local decision logic. For secure software foundations and update mechanisms (L2.1–L2.2), the main failsafes are staged rollout, rollback, secure recovery partitions, and autonomous node quarantine via eBPF network policy when attestation or update verification fails. For data-centric security (L5.1–L5.6), the relevant failsafes are TEE abstraction layers to avoid vendor lock-in during the dependency period, hybrid cryptographic schemes combining classical and PQC mechanisms for backward compatibility and harvest resistance, crypto-agility frameworks enabling rapid algorithm replacement, and diversification across PQC algorithm families, including lattice-, code-, and hash-based approaches, to avoid single-point cryptanalytic failure. For network resilience and Island Mode (L6.1–L6.4), failsafes include segmented autonomy zones, redundant local control paths, fallback routing, and validated autonomous operation for critical services when connectivity or orchestration planes are degraded.

g: D7 Transversal Meta-Layer.

Sovereignty: Across L1, L2, L5, and L6, the cluster reduces dependence on proprietary hardware roots of trust, non-European TEE stacks, classical cryptographic assumptions, and externally controlled network resilience mechanisms. Hardware sovereignty is primarily addressed through EU-designed secure elements and EUCC “High” assurance (L1.1), while cryptographic sovereignty is addressed through PQC migration, crypto-agility, and selected critical-link QKD (L5.2 and L5.5). **Trust and Ethics:** EUCC “High” aligns hardware assurance with Common Criteria EAL4+ and AVA_VAN.5 expectations; PQC and confidential-computing migration support GDPR Article 32 confidentiality and integrity obligations [28]; and the 18-Member-State joint statement “Securing Tomorrow, Today” (November 2024) formalises PQC as a collective sovereignty concern. **Sustainable Security:** Energy-efficient secure silicon and lightweight cryptography support far-edge deployment without premature device replacement, while resilient update mechanisms,

crypto-agility, and validated Island Mode extend the operational lifetime of critical infrastructure assets.

B. CLUSTER 2: COGNITIVE SECURITY AND AUTONOMOUS CYBER DEFENCE

Cluster 2 consolidates the capabilities that enable cognitive security operations and autonomous cyber defence across the continuum: software and AI lifecycle assurance (CL1.2–CL1.4), federated SOC and cyber threat intelligence functions (CL2.1 and CL2.5), high-precision telemetry and AI-native analytics (CL2.2–CL2.4), deception and active defence (CL2.6), Zero Trust reconfiguration and containment (L4.1), telecom telemetry integration (L6.1), and trustworthy algorithmic accountability for autonomous security decisions (CL4.7). The cluster is structured by three inseparable sub-dimensions: AI as a security instrument, including sovereign models, edge-native agents, autonomous SOC, and Moving Target Defence; AI as an adversary amplifier, including polymorphic malware, automated exploit discovery, and scaled social engineering; and AI as a target, where training pipelines and inference endpoints become attack surfaces under AI Act Article 9.

a: D1 Baseline.

At the operational cyber-resilience layer (CL2.1–CL2.6), European SOC and CTI capabilities remain fragmented across organisations, Member States, and commercial platforms. US-platform SIEMs dominate the EU market, with no EU-headquartered vendor in the Gartner Leaders quadrant [32], and average APT dwell times in European organisations remain approximately 150–200 days. At the AI lifecycle layer (CL1.2 and CL1.4), sovereign security-specific foundation models and standardised adversarial robustness validation for security AI remain underdeveloped; OpenEuroLLM [80] does not yet include a dedicated security stream. At the analytics and telemetry layer (CL2.2–CL2.4 and L6.1), NWDAF-enabled telecom telemetry and LLM-assisted Sec-Ops remain at pilot or early integration stage rather than being operationally federated across the continuum. At the governance and assurance layer (CL4.7), trustworthy, auditable, and AI Act-compliant autonomous security decision-making remains immature, particularly for systems that support containment, reconfiguration, or prioritisation of critical incidents.

b: D2 High-Impact Activities and Target Outcomes.

Phase 1 activities should establish the technical basis for cognitive cyber defence: integrate commercial LLMs into SOC workflows for alert triage, summarisation, and MITRE ATT&CK mapping (CL2.1 and CL2.5); route NWDAF telemetry into centralised AI analytics in three to five 5G SA pilots [69] (CL2.2–CL2.4 and L6.1); train a first EU-sovereign security-focused SLM (≤ 10 B parameters) on GDPR-compliant security corpora using EuroHPC [81] (CL1.2 and CL2.4); apply ML-based UEBA and deception-supported evidence collection for APT IoC detection (CL2.3

and CL2.6); and demonstrate sub-72 h dwell time in at least one certified NIS2-essential pilot. Phase 2 activities should scale cognitive security into operational environments: deploy EU-sovereign SLMs at the edge for real-time SOC analytics without dependence on external cloud inference (CL2.4); deploy deception infrastructure, including honeynets and credential canaries, for calibration and high-fidelity detection (CL2.6); introduce autonomous triage under ETSI GS ZSM 009 [82] with human oversight and auditability (CL2.1, CL2.4, and CL4.7); and enable Zero Trust reconfiguration and containment actions aligned with NIST SP 800-207 [4] (L4.1). The target for Phase 2 is sub-72 h dwell time across certified critical-infrastructure deployments. Phase 3 activities should consolidate autonomous cyber defence as a trusted operational capability: deploy swarm-based agentic AI for decentralised threat intelligence and coordinated response (CL2.1 and CL2.4); operate continuous Moving Target Defence across IP addressing, topology, and software diversity (CL2.6 and L4.1); and implement continuous automated adversarial robustness validation for security AI (CL1.4 and CL4.7). The target for Phase 3 is ≤ 24 h dwell time across NIS2-essential deployments and EU-sovereign AI security tools covering at least $\geq 50\%$ of new critical-infrastructure SOC deployments by 2040.

c: D3 Primary Drivers.

The primary threat drivers for this cluster arise from the accelerating use of AI by both defenders and adversaries. At the AI lifecycle and model-assurance layer (CL1.2–CL1.4), generative AI can be weaponised for polymorphic malware, automated exploit discovery, adversarial prompt manipulation, and scaled social engineering, while training pipelines and inference endpoints become attack surfaces in their own right. At the operational cyber-resilience layer (CL2.1–CL2.6), nation-state actors increasingly blend espionage with organised crime, use memory-only execution, abuse legitimate administrative tools, and exploit fragmented SOC and CTI workflows to extend dwell time. At the analytics and telecom-telemetry layer (CL2.2–CL2.4 and L6.1), high-volume telemetry, NWDAF integration, and LLM-assisted triage create both detection opportunities and new risks of automation bias, data leakage, and adversarial manipulation. At the governance and assurance layer (CL4.7), autonomous or semi-autonomous security decisions raise accountability, auditability, and conformity-assessment challenges. The regulatory drivers are correspondingly concentrated around AI and operational resilience: AI Act Articles 9, 13, and 15 create mandatory requirements for high-risk autonomous security AI [12]; NIS2 Article 21(2)(b) establishes incident-handling obligations [9]; and conformity assessment capacity is expected to lag obligations by approximately 24–36 months [17].

d: D4 Catalysts and Enablers.

The main catalysts for this cluster map to the capabilities required for cognitive security operations. For AI

lifecycle assurance and security-specific models (CL1.2–CL1.4), EuroHPC AI Factories provide the main compute and training infrastructure, with 19 confirmed sites across 16 Member States, cumulative funding above €2.7B, and IT4LIA, PIAST, and MeluXina-AI listing cybersecurity as a vertical [81]. For federated SOC, CTI, and operational response (CL2.1 and CL2.5), MITRE ATT&CK [71], STIX/TAXII 2.1 [72], [73], ENISA threat landscape reporting [70], and the ECCC Strategic Agenda [79] provide the main knowledge, coordination, and governance enablers. For AI-native analytics, NWDAF integration, and autonomous triage (CL2.2–CL2.4 and L6.1), ETSI GS ZSM 009 [82], telecom analytics interfaces, and 5G/6G testbeds provide the operational automation base. For trustworthy autonomous security decisions (CL4.7), the NIST AI RMF and Generative AI Profile [64], [65], together with AI Act conformity-assessment infrastructure, provide the main assurance framework.

e: D5 Disruptors and Systemic Risks.

The main systemic risks affect both the AI components and the cyber-defence workflows that depend on them. For AI lifecycle assurance (CL1.2–CL1.4), hallucinations, adversarial prompt manipulation, data leakage into models, poisoned training data, and the absence of a standardised adversarial robustness test suite for security AI could undermine trust in autonomous or semi-autonomous tools. For operational cyber resilience (CL2.1–CL2.6), intelligence fragmentation, weak information-sharing incentives, and adversarial adaptation outpacing defensive innovation could prevent federated SOCs from achieving the intended dwell-time reduction. For analytics and telecom telemetry (CL2.2–CL2.4 and L6.1), high-volume data flows may increase false positives, automation bias, and dependence on opaque model outputs unless explainability and auditability are built into triage and response workflows. For trustworthy AI governance (CL4.7), conformity-assessment body capacity shortfalls [17] could delay deployment of compliant autonomous security agents even where the technical capabilities are mature.

f: D6 Failsafe Mechanisms.

Failsafe mechanisms should bound autonomy and preserve operational control when AI-assisted security systems behave unpredictably or lack sufficient assurance. For AI lifecycle capabilities (CL1.2–CL1.4), failsafes include adversarial robustness testing, red-teaming, model rollback, dataset provenance checks, and shadow-mode operation before production deployment. For operational cyber-resilience capabilities (CL2.1–CL2.6), layered detection, deception, rapid recovery playbooks, and human-in-the-loop validation for Tier-1 critical actions should prevent over-reliance on automated triage. For autonomous analytics and orchestration (CL2.4, L4.1, and L6.1), bounded autonomy governed by ETSI GS ZSM 009 [82], policy-constrained orchestration, audit logging, and shadow-mode rollback should limit unstable control loops. For Zero Trust containment and accountable

decision-making (L4.1 and CL4.7), autonomous real-time micro-segmentation should quarantine compromised components only within predefined policy boundaries, with traceable justification and escalation paths.

g: D7 Transversal Meta-Layer.

Sovereignty: Across CL1, CL2, CL4.7, L4.1, and L6.1, the cluster reduces dependence on non-European SIEM platforms, external foundation-model providers, and opaque cloud-based inference for security operations by promoting EU-sovereign security models, edge-native analytics, federated SOC coordination, and auditable autonomous response. **Trust and Ethics:** AI Act Article 9 compliance-by-design, explainability, auditability, and accountability are central for autonomous or semi-autonomous security decisions, while NIS2 sector-specific implementing acts should reference structured CTI frameworks such as MITRE ATT&CK [71] to improve operational comparability across sectors. **Sustainable Security:** Edge-native SLMs and localised analytics can reduce inference-time energy costs and latency relative to repeated cloud round-trips for high-volume telemetry, while automation can reduce analyst overload if bounded by human oversight, robust testing, and transparent governance.

C. CLUSTER 3: FEDERATED OPERATIONAL SOVEREIGNTY

Cluster 3 consolidates the capabilities that transform sovereign components into a collectively sovereign continuum: heterogeneous interoperability and Simpl integration (CL3.1), neutral-host and multi-tenant isolation (CL3.2), pan-European NaaS federation and cross-border edge roaming (CL3.3), semantic security interoperability (CL3.4), cyber range and digital twin federation (CL3.5), dynamic supply-chain verification and Runtime SBOMs (CL1.3), continuous automated certification and assurance (CL4.2), privacy-preserving federated computation (CL4.4), and Zero Trust policy enforcement across shared infrastructure (L4.1). Supply-chain verification bridges Clusters 1 and 3 because software, runtime, and federation assurance increasingly depend on the hardware roots of trust and attestation capabilities matured in Cluster 1.

a: D1 Baseline.

At the interoperability layer (CL3.1–CL3.4), European cloud-edge ecosystems remain fragmented across proprietary vendor silos, incompatible security policies, and uneven semantic models. The Simpl-Open iterative MVP has been available since January 2025, with enhanced releases in June and December 2025 [58], but OASC MIM6 (Security management) remains a “Work item” without a formally published specification. At the supply-chain and runtime-verification layer (CL1.3 and CL4.2), current supply chains remain opaque and globally distributed: CRA Annex I Part II mandates static SBOM disclosure from December 2027 [10], NIS2 Article 21(2)(h) mandates supply-chain risk assessment [9], and DORA Article 28 requires continuous ICT

third-party monitoring [11], operationalised through the first 19 Critical ICT Third-Party Provider designations on 18 November 2025 [83]. At the federation and connectivity layer (CL3.3, L6.1, and L4.1), the CAMARA Fall 2025 meta-release comprises 60 APIs, of which 10 are stable [74], and GSMA Open Gateway reaches 86 operator groups [84], but cross-border edge compute roaming does not yet exist at operational scale. At the governance and assurance layer (CL4.2–CL4.6), continuous assurance, cyber risk quantification, and policy portability remain insufficiently integrated across providers and jurisdictions.

b: D2 High-Impact Activities and Target Outcomes.

Phase 1 activities should establish the technical basis for federated operational sovereignty: run Simpl security MIM pilots in three Common European Data Spaces, including Manufacturing, Health [85], and Energy (CL3.1 and CL3.4); deploy harmonised CAMARA APIs in two to three bilateral 5G corridor agreements (CL3.3 and L6.1); submit CRA-mandated static SBOM declarations and initiate Digital Product Passport pilots (CL1.3 and CL4.2); advance zero-knowledge proof verification towards demonstrator readiness for supply-chain attestation (CL1.3 and CL4.2); and validate Zero Trust isolation policies for shared and multi-tenant infrastructure (CL3.2 and L4.1). Phase 2 activities should scale federation into operational environments: automate policy translation via standardised Simpl MIMs across active Data Spaces (CL3.1 and CL3.4); deploy Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure in at least ≥ 5 Member States (CL1.3 and CL4.2); implement automatic quarantine of unverified components in isolated network slices (CL3.2, L4.1, and L6.1); scale CAMARA cross-border roaming to at least ≥ 5 operator pairs (CL3.3); and operationalise “Data Visa” mechanisms for dynamic AI model migration across jurisdictions (CL3.1, CL4.4, and CL4.7). The target for Phase 2 is standardised integration of Simpl-based security interoperability, runtime supply-chain attestation, and cross-border edge federation across multiple Member States. Phase 3 activities should consolidate federated operational sovereignty as the default model: establish Simpl-based security as the universal Data Space standard (CL3.1 and CL3.4); operate ZKP-verified supply-chain attestation at real-time transaction scale (CL1.3 and CL4.2); deploy an operational EU NaaS federation for cross-border edge and connectivity services (CL3.3 and L6.1); and enact a “Schengen for Cloud” legal framework resolving US CLOUD Act jurisdictional conflict. The target for Phase 3 is a unified European trust fabric in which Simpl MIMs are ratified as CEN/CENELEC standards and referenced in the EU Cloud Rulebook, non-EU hyperscaler market share in new EU regulated cloud procurement falls below 40%, and federated policy, supply-chain verification, and NaaS roaming operate at commercial scale by 2040.

c: D3 Primary Drivers.

The primary threat drivers for this cluster arise from the difficulty of federating security across heterogeneous providers, jurisdictions, and supply chains. At the interoperability and federation layer (CL3.1–CL3.4), proprietary security policies, incompatible APIs, and uneven semantic models create fragmentation that prevents European operators from scaling against US and Chinese hyperscalers. At the supply-chain and runtime-verification layer (CL1.3 and CL4.2), SolarWinds-class software supply-chain attacks demonstrate the risk of trusting static artefacts, opaque build pipelines, and unverifiable runtime components. At the shared-infrastructure and network-federation layer (CL3.2, CL3.3, L4.1, and L6.1), disaggregated telco clouds and cross-border edge deployments introduce multi-vendor complexity, tenant-isolation risks, and policy-enforcement gaps. The regulatory drivers are correspondingly distributed across the underlying capabilities: the Data Act, applicable from 12 September 2025, strengthens portability and switching obligations [27]; NIS2 requires coordinated incident response and supply-chain risk management [9]; the CRA introduces SBOM obligations from December 2027 [10]; DORA requires continuous ICT third-party monitoring [11]; and the Digital Networks Act proposal (COM(2026) 16, January 2026) signals European Commission intent for network federation, although it does not mandate API standards.

d: D4 Catalysts and Enablers.

The main catalysts for this cluster map to the capabilities needed for federated operational sovereignty. For heterogeneous interoperability and Simpl integration (CL3.1 and CL3.4), Digital Europe Programme funding for Simpl [58], CEN-CLC/JTC 25 data-space standardisation, the Eclipse Dataspace Protocol, and OASC/NGSI-LD semantic interoperability provide the main standardisation and implementation base. For pan-European NaaS federation and cross-border edge roaming (CL3.3 and L6.1), SNS JU activities [57], CAMARA APIs, and emerging aggregator models such as Aduna provide the main technical and ecosystem enablers. For supply-chain verification and continuous assurance (CL1.3 and CL4.2), the IPCEI-CIS ICRA Reference Architecture v2.0 (January 2026, total €2.6B, 19 participants, 12 Member States) [86], the EU Ecodesign Regulation Digital Product Passport, and DORA Delegated Regulation (EU) 2024/1502 for CTPP assessment [11] provide the principal policy and assurance instruments. For Zero Trust isolation and shared-infrastructure governance (CL3.2 and L4.1), reusable policy profiles, certification evidence exchange, and multi-tenant isolation pilots are the key implementation enablers.

e: D5 Disruptors and Systemic Risks.

The main systemic risks affect different parts of the federation stack. For interoperability and semantic security (CL3.1 and CL3.4), vendor resistance to open standards, proprietary API ecosystems, and uneven Member State compliance ma-

turity could prevent convergence on common security mechanisms. For neutral-host and multi-tenant isolation (CL3.2 and L4.1), weakest-link failures in federated systems could allow one poorly governed participant to undermine the trust assumptions of the wider federation. For supply-chain verification and continuous assurance (CL1.3 and CL4.2), limited multi-tier auditability, concentration of manufacturing and software dependencies, and SME exclusion from compliance pathways could weaken adoption. For NaaS federation and cross-border edge roaming (CL3.3 and L6.1), hyperscaler platform dominance through proprietary API ecosystems, with approximately 70% EU cloud share [7], could limit European scale-up. Across all federation capabilities, unresolved US CLOUD Act jurisdictional conflict [29] remains a structural sovereignty risk.

f: D6 Failsafe Mechanisms.

Failsafe mechanisms should preserve trust when federation assumptions fail or when some providers cannot meet the shared baseline. For heterogeneous interoperability and Simpl-based federation (CL3.1 and CL3.4), failsafes include autonomous unfederation, policy rollback, and fallback to minimum common security profiles. For neutral-host and multi-tenant isolation (CL3.2 and L4.1), Zero Trust architectures should ensure that federation does not imply implicit trust [4], with cryptographic attestation of policy enforcement and graceful isolation of unverified components into quarantined network slices. For supply-chain verification and continuous assurance (CL1.3 and CL4.2), runtime attestation, trusted fallback components, and multi-vendor diversification reduce dependency on any single provider or assurance path. For NaaS federation and network resilience (CL3.3 and L6.1), graceful degradation should redirect workloads to national infrastructure, regional autonomy zones, or multi-operator fallback routing when cross-border federation is unavailable or legally constrained.

g: D7 Transversal Meta-Layer.

Sovereignty: Across CL3, CL1.3, CL4.2, L4.1, and L6.1, the cluster reduces hyperscaler lock-in by making interoperability, runtime assurance, policy portability, and cross-border edge federation enforceable rather than aspirational, while also addressing structural CLOUD Act conflict. **Trust and Ethics:** Transparent and verifiable federation mechanisms support cross-border trust by making policy enforcement, runtime integrity, and assurance evidence auditable across providers and jurisdictions; the 2025 DINUM Bilan reports 70% European-provider share in French public cloud procurement market-wide and 99% within the State perimeter [87]. **Sustainable Security:** Shared federated infrastructure, reusable certification evidence, and common security profiles reduce duplication of compute, audit, and certification overhead across Member States while supporting SME access to compliant cloud-edge services.

D. CROSS-CUTTING ENABLERS

Four dependencies span at least two clusters and require dedicated governance to avoid becoming systemic bottlenecks.

EU Chips Act sovereign fabrication (Clusters 1 and 3). Five Chips JU pilot lines are operational, with combined EU and national funding of €3.7B [77]. No pilot line is dedicated to secure element fabrication; restriction of access to any major non-EU foundry would immediately impair EU critical infrastructure attestation capability. Mitigation: strategic stockpiling, second-source agreements, and dedicated pilot lines for security-critical components.

AI Act conformity assessment infrastructure (Cluster 2). A 24–36 month gap between AI Act obligations and CA body operational readiness is documented [17]. As of March 2026, 8 of 27 Member States have designated single points of contact. Mitigation: accelerated CEN/CENELEC adversarial robustness test suite development and AI Act sandbox provisions for security use cases.

Runtime SBOM attestation (Clusters 1 and 3). Static SBOMs (CRA-mandated) are insufficient for dynamic continuum operations. Progression to cryptographic proof-of-execution and ZKP batch verification is a shared Phase 2–3 transition condition.

SME access and inclusion (all clusters). Cost barriers exclude SMEs from sovereign hardware alternatives, AI-powered SOC tooling, and Simpl MIM compliance. Digital Europe Programme co-funding should target shared-infrastructure pathways and subsidised certification programmes.

VI. STAGE 3: PHASE LOGIC AND TRANSITION MATRICES

VII. STRATEGIC RECOMMENDATIONS

The 2040 destination is a secure, sovereign, interoperable, and resilient European computing continuum. Six actions follow from the analysis.

Sovereign secure silicon. Combine Horizon Europe support with the EU Chips Act to advance open-source RISC-V secure elements and pilot fabrication lines reaching EUCC “High” assurance. The 2025 certifications achieved by Infineon, NXP, and STMicroelectronics [30] demonstrate framework readiness; the remaining gap is open-source, EU-controlled fabrication.

Operationalised PQC migration. Deliver crypto-agility middleware for edge and IoT aligned with the NIS CG PQC roadmap milestones [8] and ETSI TS 103 744 hybrid key exchange profiles [53]. Make hybrid cryptographic deployment the default for new critical infrastructure.

Security interoperability through Simpl. Move OASC MIM6 (Security management) from Work item status to a published specification, then to a ratified CEN/CENELEC standard referenced in the EU Cloud Rulebook, so that interoperability carries procurement-level enforcement.

Continuous certification. Use EUCC [14] as the foundation and unblock EUCS [13], plausibly through the 2026

Table 2. Cluster 1 transition matrix, Sovereign Trusted Infrastructure.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Applied R&D to Fragmented Piloting	Standardised Integration	Ubiquitous Operation
SSL	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	RISC-V secure element pilots in three to five Horizon Europe consortia; Intel/AMD/ARM TEEs dominate all operational deployments; classical cryptography on far-edge IoT; Island Mode validated in 5G testbeds by SNS JU consortia.	First EUCC “High” EU-manufactured RISC-V secure elements in production from Chips JU pilot lines; deployed in energy substations and telecom critical infrastructure in ≥ 5 Member States; RISC-V Keystone TEEs operational in multi-provider Data Spaces alongside Intel TDX; 72 h Island Mode certified in cross-border 5G corridors.	EU-designed secure elements as default in new EU ICT products; $\geq 60\%$ of critical infrastructure edge nodes on EUCC “High” EU-designed elements; FHE end-to-end across continuum; 72 h autonomous Island Mode mandatory in NIS2-essential and DORA-critical deployments.
Critical Enabler	EUCC “High” certification of an EU-origin open-source RISC-V secure element before end of phase.	Ratification of ETSI TS profiles for FIPS 203/204/205 tailored to EU deployment contexts; operational Chips JU fabrication capacity at the advanced nodes required for secure elements.	Mandatory EUCC “High” without waiver for new EU ICT products; PQC-native silicon as EU procurement default.

Table 3. Cluster 2 transition matrix, Cognitive Security and Autonomous Cyber Defence.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Applied R&D, Siloed AI Copilots	Standardised Integration	Ubiquitous Operation
SSL	High Reliance on US Foundation Models	Sovereign EU Edge LLMs Emerging	Global Leadership in Trustworthy AI Security
Operational State	Commercial LLMs integrated into SOC workflows for triage; NWDAF pilots in three to five 5G SA deployments; first EU-sovereign SLM under training on EuroHPC AI Factory compute; sub-72 h dwell time achieved in at least one certified pilot; no operational autonomous containment agent.	EU-sovereign SLMs deployed at the edge for real-time SOC analytics without cloud inference; autonomous triage governed by ETSI GS ZSM 009 in production; sub-72 h dwell time in certified critical infrastructure deployments; deception infrastructure operational for calibration.	Swarm-based agentic AI defending the continuum via self-evolving decentralised CTI; Moving Target Defence continuous in certified environments; ≤ 24 h dwell time across NIS2-essential deployments; EU-sovereign AI security tools in $\geq 50\%$ of new critical infrastructure SOC deployments.
Critical Enabler	CEN/CENELEC or ETSI TC CYBER publication of a standardised adversarial robustness test suite for security AI before end of phase.	Operationalisation of an AI Act Article 9 conformity assessment framework specific to autonomous security agents; activation of EuroHPC AI Factory compute contracts for EU-sovereign security model training.	International adoption of EU AI Act frameworks for trustworthy autonomous security agents by at least one non-EU partner jurisdiction.

Cybersecurity Act revision (COM(2026) 13) [16], to extend continuous assurance to cloud services.

Federated SOC and CTI. Build on DORA CTPP oversight [83] and cross-border CSIRT cooperation [68] to support pan-European situational awareness.

SME inclusion mandate. Implement tiered certification, shared cyber range access, and reusable compliance tooling so that NIS2, CRA, and AI Act compliance does not become a structural barrier to innovation.

VIII. DISCUSSION

A. VALIDATION

The roadmap is treated as an evolving strategic artefact benefiting from expert review, stakeholder consultation, and iterative revision. Validation is conducted through expert workshops within the NexusForum and EuCloudEdgeIoT Working Group on Cybersecurity; consultation with national cybersecurity agencies and industry; and policy dialogue with European Commission services. Validation addresses completeness and relevance of the architectural decomposition, clarity and non-overlap of capability and priority formulations, coherence of priority clustering, credibility

of transition conditions, and reasonableness of Sovereignty Level assessments.

B. DISCUSSIONS, LIMITATIONS AND FUTURE WORK

The methodology is intended to support strategic agenda setting, expert discussion, cross-priority comparison, and iterative roadmap refinement under conditions of ecosystem uncertainty. Its value lies in providing a shared analytical structure through which stakeholders can align research, policy, deployment, and investment decisions around a common set of objectives.

At the same time, the roadmap should not be interpreted as a precise forecast of future cybersecurity developments. The Cloud–Edge–IoT landscape is shaped by rapidly evolving technologies, regulatory implementation, market dynamics, and geopolitical conditions, all of which may alter the trajectory of the priority clusters. The analysis therefore offers a strategic synthesis of the state of the art and expert judgement, rather than a deterministic prediction of how the cybersecurity landscape will evolve.

Future work could extend the roadmap in three directions. First, the architectural analysis could be instantiated in

Table 4. Cluster 3 transition matrix, Federated Operational Sovereignty.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Fragmented Piloting	Standardised Integration	Ubiquitous Operation
SSL	Mixed: Emerging EU Ecosystem / Hyper-scaler Dominance	Emerging EU Ecosystem	Full Federated Sovereign Autonomy
Operational State	Simpl security MIM pilots in three Common European Data Spaces with manual policy coordination; CAMARA APIs in two to three bilateral 5G corridors; no operational cross-border edge compute roaming; CRA-mandated static SBOMs submitted as point-in-time artefacts; DPP pilots initiated; hyperscalers retain approximately 70% of EU cloud market.	Automated policy translation via standardised Simpl MIMs across active Data Spaces; Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure in ≥ 5 Member States; ZKP batch verification operational; CAMARA roaming across ≥ 5 operator pairs; “Data Visa” mechanisms operational; non-hyperscaler share $\geq 30\%$ in new regulated-sector procurement.	Simpl-based security as universal cross-provider Data Space standard; ZKP-verified supply chain attestation at real-time scale; operational EU NaaS federation; “Schengen for Cloud” framework enacted; non-EU hyperscaler share below 40% in EU regulated cloud procurement for new deployments; EU supply chain standards adopted in ISO/IEC and by at least two partner jurisdictions.
Critical Enabler	Simpl-Open iterative MVP consolidation with OASC MIM6 (Security management) progressing from Work item to formally published specification.	Ratification of Simpl security MIMs as CEN/CENELEC standards referenced as mandatory in the EU Cloud Rulebook; completion of DORA continuous ICT third-party monitoring operationalisation.	Enactment of the “Schengen for Cloud” legal framework resolving US CLOUD Act jurisdictional conflict.

sector-specific use cases, such as energy, healthcare, finance, and public procurement, to examine how cybersecurity priorities are affected by sector-specific regulation, procurement rules, supply-chain constraints, and market structure. This would help capture dependencies that fall outside a cybersecurity-centred perspective. Second, the Sovereignty Level construct should be further operationalised by identifying measurable proxies for strategic dependency, such as provider concentration, certification availability, fabrication capacity, and cross-border deployability. Third, the roadmap should be maintained as an iterative and dynamic instrument, periodically revised as new empirical evidence, stakeholder feedback, regulatory milestones, and technological developments become available.

IX. CONCLUSION

The Cloud-Edge-IoT continuum requires Europe to move beyond reactive and fragmented cybersecurity models towards autonomous, federated, and certified Security-by-Design. The structured methodology presented here identifies security dimensions, derives strategic capabilities, and consolidates them into priority clusters aligned with resilience and digital sovereignty objectives. The ten priorities form an interdependent strategic portfolio that must be developed in parallel. The 2025–2026 period has produced concrete milestones (EUCC “High” certifications across all three major EU secure element manufacturers, a binding PQC migration framework, DORA operationalisation through CTPP designation, and emerging Simpl infrastructure) and persistent gaps (NIS2 transposition delays, EUCS deadlock, contested semiconductor targets, continued hyperscaler dominance at approximately 70% of the EU cloud market). The ambition–implementation gap defines where the coordination challenge is most acute and where strategic investment will yield the greatest returns. Future work should extend this analysis through empirical validation of Sovereignty Level assessments, sector-specific instantiation (energy, healthcare,

financial services), and integration with the next EU Framework Programme funding landscape.

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